

Question

M is a metallic element, **A** is a 1:1 type salt containing **M**, with a molar mass of 223.0 g/mol. When BF_3 is passed into an anhydrous hydrogen fluoride solution of **A**, an diamagnetic red solid **B** is obtained, and a 1 : 1 type salt **C** precipitates simultaneously. 0.1227 g of **A** is dissolved in anhydrous hydrogen fluoride, and allowed to react fully under a 2 atm atmosphere of Xe, yielding 0.1862 g of solid, the components of which are a mixture of white solid **D**, dark brown paramagnetic solid **E**, and $2\text{KF} \cdot 5\text{HF}$. **B** can also react directly with Xe gas to obtain **D** and **E**. **E** has extremely strong oxidizing power, for example, when **E** is added to NaOH solution, it decomposes violently, producing a dark brown precipitate **F** and releasing gas; **E** reacts with chloroform to produce **G** (also known as Freon-11) and yellow solid **H**, etc.; **H** is easily soluble in water (1800 g/L). When **H** is dissolved in anhydrous hydrogen fluoride, and colloidal **M** element is added, a brown solid **I** precipitates, **I** has a cadmium iodide inverse structure; **I** undergoes disproportionation upon contact with water, yielding **M** element and solid **H** again.

It is known that **B** exhibits a chain structure.

The following statements are correct:

- A.** All other options are incorrect
- B.** **M** The ground state has two single electrons.
- C.** The constituent elements of **A** include two short-period elements.
- D.** The number of binary compounds in **A** — **I** is greater than 6.
- E.** **A** — **I** contains a compound composed of four elements.
- F.** **A** — **I** all contain the same short-period element.

G. The chemical equation for the decomposition of **E** in NaOH, balanced with the simplest coefficient ratio, has a greatest common divisor of 3 between the sum of the product coefficients and the sum of the reactant coefficients.

H. The d_{xy} orbital of the M element in **B** is in a half-filled state.

I. The $d_{x^2+y^2}$ orbital of element **M** in **B** is fully occupied.

Answer

Correct Answer: **G**

Detailed Explanation

The starting point of this question is that **H** is easily soluble in water (1800 g/L). According to element knowledge, it can be immediately concluded that **H** is AgF.

CHECKPOINT

1 PTS

Solubility information indicates that **H** is AgF

Therefore, **M** is Ag.

CHECKPOINT

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M is Ag

G is also known as Freon-11, then **G** is CFCl_3 , and **E** contains F;

CHECKPOINT

1 PTS

G is CFCl_3

A is soluble in anhydrous hydrogen fluoride and reacts with Xe to produce $2\text{KF} \cdot 5\text{HF}$, then **A** contains K and Ag, and most likely contains F. Since it is a 1 : 1 type salt, the remaining relative molecular mass is $223.0 - 107.9 - 39.1 = 76.0 = 19.0 \times 4$, which is four fluorine atoms. Thus, the chemical formula of **A** is KAgF_4 .

CHECKPOINT

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The chemical formula of **A** is KAgF_4

B is diamagnetic, then **B** contains d^8 configuration of Ag^{3+} ; therefore, **B** is AgF_3 . The 1:1 type salt **C** must contain K. Since BF_3 is added, **C** is KBF_4 .

CHECKPOINT

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B is AgF_3

CHECKPOINT

1 PTS

C is KBF_4

A is soluble in anhydrous hydrogen fluoride and reacts with Xe. Since Ag(III) has strong oxidizing properties, KAgF_4 is a strong fluorinating agent and can oxidize Xe. The white solid **D** should be XeF_2 . Ag(III) is reduced to a paramagnetic species, which can only be d^9 configuration of Ag^{2+} , so **E** is AgF_2 .

CHECKPOINT

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D is XeF_2

CHECKPOINT

1 PTS

Paramagnetic species, can only be d^9 configuration of Ag^{2+}

CHECKPOINT

1 PTS

E is AgF_2

AgF_2 decomposes in NaOH solution. Ag(II) is also a highly unstable valence state, so it can oxidize water to produce oxygen, and itself is reduced. The dark brown precipitate **F** is obviously Ag_2O .

CHECKPOINT

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F is Ag_2O

AgF reacts with Ag , which can only be a comproportionation reaction. Associating with the structure of trans-CdI_2 , the generated **I** should be Ag_2F .

CHECKPOINT

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I is Ag_2F

Thus, determine the options:

The ground state configuration of Ag is $4d^{10}5s^1$, with one single electron, so B is incorrect.

KAgF_4 only has one short-period element, F, so C is incorrect.

According to the chemical formula, options D, E, and F are all incorrect.

AgF_2 decomposes to produce Ag_2O and oxygen, and the equation can be written as:



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The sum of the product coefficients is 15, and the sum of the reactant coefficients is 12. The greatest common divisor is 3, so option G is correct.

AgF_3 has a chain structure. It can be seen that Ag(III) in AgF_3 is coordinated by four F in a square planar configuration, and the AgF_4 units are connected by sharing F. Ag(III) is now in a square planar field. Under the d^8 configuration, $d_{x^2+y^2}$ has no electrons, and d_{xy} is fully filled, so options H and I are incorrect.

CHECKPOINT

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Ag(III) in AgF_3 is coordinated by four F in a square planar configuration

CHECKPOINT

1 PTS

In a square planar field, under the d^8 configuration, $d_{x^2+y^2}$ has no electrons, and d_{xy} is fully filled